

University of the Punjab

(Gujranwala Campus)

ARTIFICIAL INTELLIGENCE

LAB TASK 03

Submitted by:

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ROLL NO 33

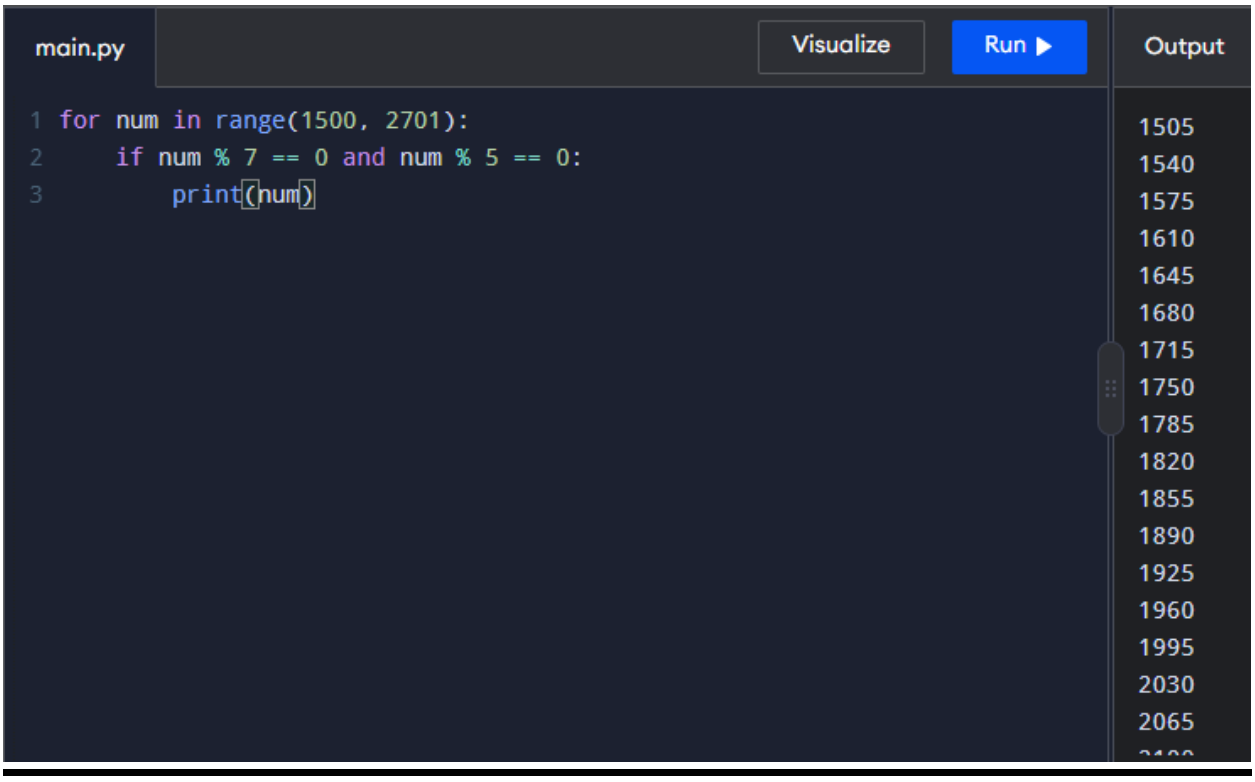
BSIT (5th Semester)

Submitted to:

Sir Ghulam Mustafa

PROGRAM 01

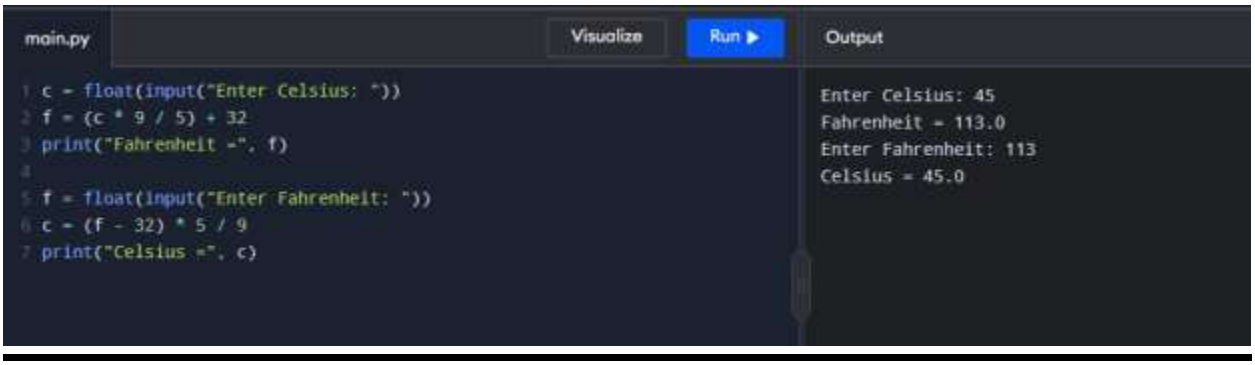
1. Write a Python program to find those numbers which are divisible by 7 and
   multiple of 5, between 1500 and 2700 (both included).



PROGRAM 02

2. Write a Python program to convert temperatures to and from Celsius, Fahrenheit.
[Formula: c/5 = f-32/9 [where c = temperature in Celsius and f = temperature in
Fahrenheit]]

Expected Output:
60°C is 140 in Fahrenheit 45°F is 7 in Celsius



PROGRAM 03

3. Write a Python program to guess a number between 1 to 9.
Note: User is prompted to enter a guess. If the user guesses wrong then the prompt
appears again until the guess is correct, on successful guess, user

will get a "Well
guessed!" message, and the program will exit.





POGRAM 04

4. Write a Python program to construct the following pattern, using a nested for loop.

```
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
```

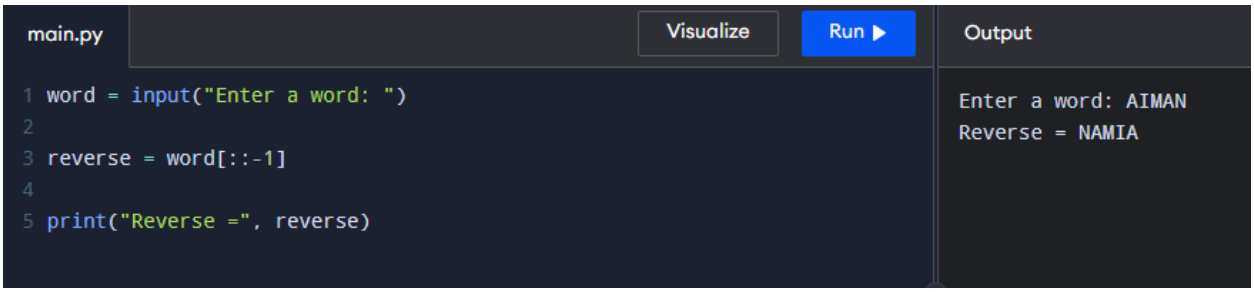
main.py | Visualize | Run ▶ | Output

```
for i in range(1, 6):
    for j in range(i):
        print("*", end=" ")
    print()

for i in range(4, 0, -1):
    for j in range(i):
        print("*", end=" ")
    print()
```

PROGRAM 05

5. Write a Python program that accepts a word from the user and reverse it.



PROGRAM 06

6. Write a Python program to count the number of even and odd numbers from a
series of numbers.

Sample numbers: numbers = (1, 2, 3, 4, 5, 6, 7, 8, 9)
Expected Output:
Number of even numbers: 5 Number of odd numbers: 4



PROGRAM 07

7. Write a Python program that prints each item and its corresponding type from the

following list.

Sample List: datalist = [1452, 11.23, 1+2j, True, 'w3resource', (0, -1), [5, 12],
{"class":"V", "section":"A"}]

```
main.py                                   Visualize   Run ▶   Output

datalist = [1452, 11.23, 1+2j, True, 'w3resource', (0, -1), [5, 12],
{"class":"V", "section":"A"}]

for item in datalist:
    print(item, type(item))
```

```
1452 <class 'int'>
11.23 <class 'float'>
(1+2j) <class 'complex'>
True <class 'bool'>
w3resource <class 'str'>
(0, -1) <class 'tuple'>
[5, 12] <class 'list'>
{'class': 'V', 'section': 'A'} <class 'dict'>
```

PROGRAM 08

8. Write a Python program that prints all the numbers from 0 to 6 except 3 and 6.
Note: Use 'continue' statement.

```
main.py                                   Visualize   Run ▶   Output

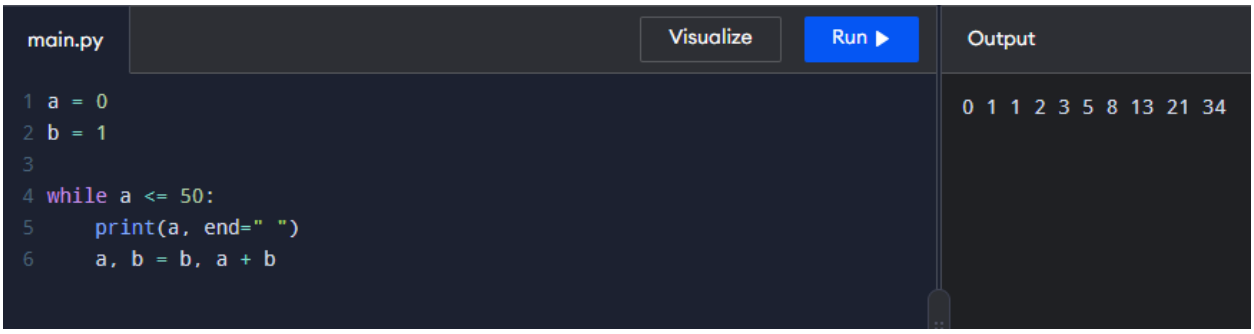
for i in range(7):
    if i == 3 or i == 6:
        continue

    print(i)
```

```
0
1
2
4
5
```

PROGRAM 09

9. Write a Python program to get the Fibonacci series between 0 to 50. Note: The
Fibonacci Sequence is the series of numbers:
0, 1, 1, 2, 3, 5, 8, 13, 21, ...
Every next number is found by adding up the two numbers before it.
Expected Output: 1 1 2 3 5 8 13 21 34



PROGRAM 10

Write a Python program which iterates the integers from 1 to 50. For multiples of
three print "Fizz" instead of the number and for the multiples of five print "Buzz".
For numbers which are multiples of both three and five print "FizzBuzz".

Sample Output: fizzbuzz

1
2
fizz 4
buzz

```python
for i in range(1, 51):

    if i % 3 == 0 and i % 5 == 0:
        print("FizzBuzz")

    elif i % 3 == 0:
        print("Fizz")

    elif i % 5 == 0:
        print("Buzz")

    else:
        print(i)
```

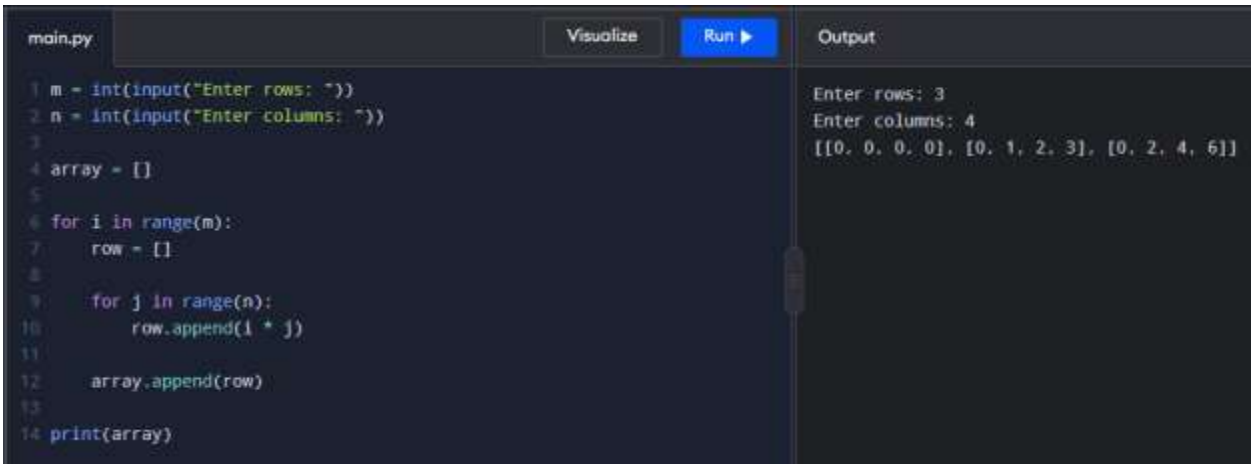
```
1
2
Fizz
4
Buzz
Fizz
7
8
Fizz
Buzz
11
Fizz
13
14
FizzBuzz
16
17
```

PROGRAM 11

. Write a Python program which takes two digits m (row) and n (column) as input
and generate a two-dimensional array. The element value in the i-th row and j-th
column of the array should be i*j.

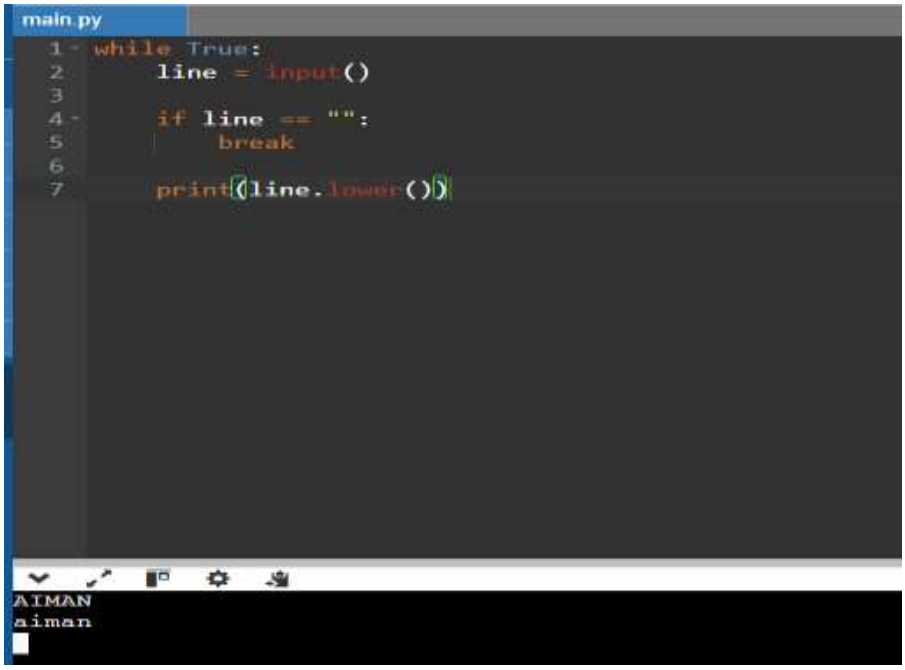
Note:
i = 0,1,.., m-1
j = 0,1, n-1.

Test Data: Rows = 3, Columns = 4
Expected Result: [[0, 0, 0, 0], [0, 1, 2, 3], [0, 2, 4, 6]]



PROGRAM 12

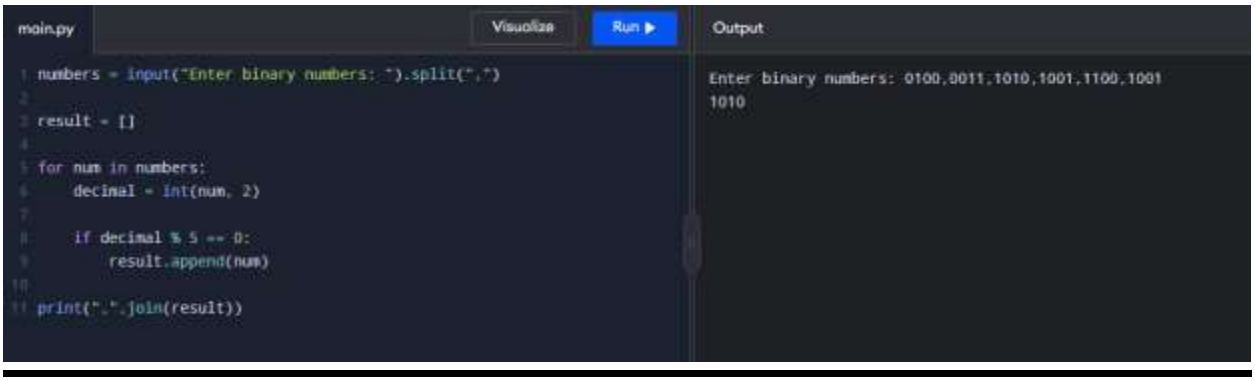
Write a Python program that accepts a sequence of lines (blank line to terminate)
as input and prints the lines as output (all characters in lower case).



PROGRAM 13

Write a Python program which accepts a sequence of comma separated 4 digit
binary numbers as its input and print the numbers that are divisible by 5 in a
comma separated sequence.

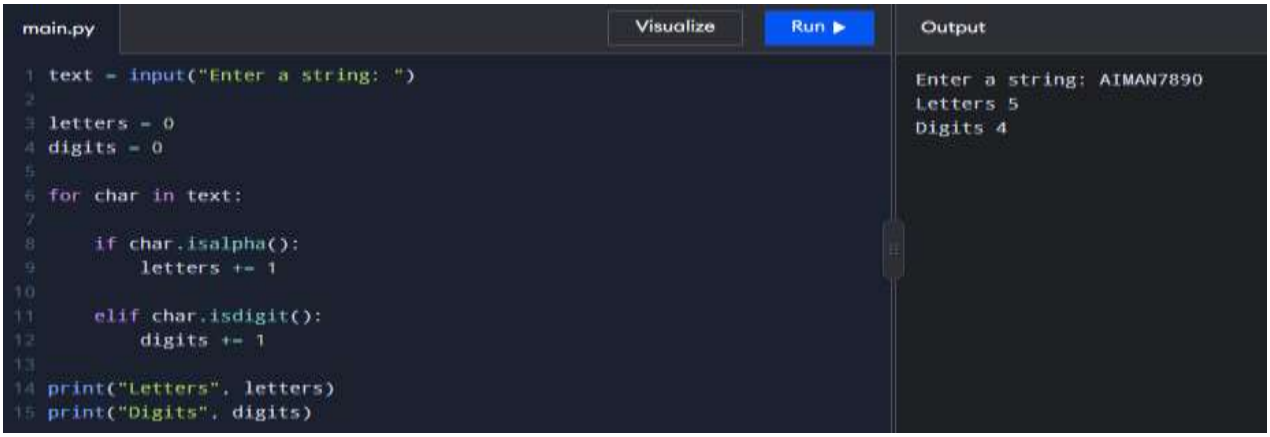
Sample Data: 0100,0011,1010,1001,1100,1001
Expected Output: 1010



PROGRAM 14

Write a Python program that accepts a string and calculate the number of digits
and letters.

Sample Data: Python 3.2 Expected Output:
Letters 6
Digits 2



PROGRAM 15

14. Write a Python program to check the validity of password input by users. Validation:

- At least 1 letter between [a-z] and 1 letter between [A-Z].
- At least 1 number between [0-9].
- At least 1 character from [$#@].
- Minimum length 6 characters.
- Maximum length 16 characters.

```python
password = input("Enter password: ")

lower = 0
upper = 0
digit = 0
special = 0

for char in password:

    if char.islower():
        lower = 1

    elif char.isupper():
        upper = 1

    elif char.isdigit():
        digit = 1
```

```python
for char in password:

    if char.islower():
        lower = 1

    elif char.isupper():
        upper = 1

    elif char.isdigit():
        digit = 1

    elif char in "$#@":
        special = 1

if 6 <= len(password) <= 16 and lower and upper and digit and special:
    print("Valid Password")
else:
    print("Invalid Password")
```

```
Enter password: ABc@123
Valid Password
```

```
for char in password:

    if char.islower():
        lower = 1

    elif char.isupper():
        upper = 1

    elif char.isdigit():
        digit = 1

    elif char in "$#@":
        special = 1

if 6 <= len(password) <= 16 and lower and upper and digit and special:
    print("Valid Password")
else:
    print("Invalid Password")
```

```
Enter password: abc098
Invalid Password
```